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ORIGINAL ARTICLE

Case Control Study

Effects of electroacupuncture combined with flurbiprofen axetil on postoperative pain and early functional recovery after breast cancer surgery

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Abstract

BACKGROUND

Breast cancer surgery often leads to acute postoperative pain that affects early recovery. Electroacupuncture (EA) is a potential adjunct to analgesics such as flurbiprofen axetil (FA), but evidence on its perioperative benefits remains limited.

AIM

To investigate the effects of EA combined with FA on postoperative pain and early functional recovery after breast cancer surgery.

METHODS

A total of 130 patients with breast cancer were randomly divided into EA group (n = 65) and control group (n = 65). Patients received EA or sham acupuncture from 30 minutes before anesthesia induction until surgery end. Both groups received FA postoperatively. We observed the occurrence of postoperative pain (incidence of moderate-to-severe pain during movement and at rest, Visual Analog Scale score) at 24 hours, 48 hours, 72 hours postoperatively, the score of basic activities of daily living (BADL) at 24 hours postoperatively and complications after surgery in both groups of patients.

RESULTS

Compared with the control group, the EA group lowered the incidence of moderate-to-severe movement-evoked pain at 24 hours postoperatively (P < 0.05), and reduced Visual Analog Scale scores during movement and at rest 24 hours after surgery (P < 0.01). The scores of BADL for grooming, dressing, toileting, and transferring (moving in and out of a bed or chair) were significantly better in the

EA group than in the control group 24 hours after surgery ($P < 0.05$), the total BADL score was significantly higher ($P < 0.01$). The incidence of postoperative complications (nausea, vomiting, loss of appetite) was significantly reduced in the EA group ($P < 0.05$).

CONCLUSION

EA combined with flurbiprofen FA can reduce the occurrence of postoperative pain and complications, improve early postoperative functional recovery after breast cancer surgery.

Key Words: Electroacupuncture; Breast cancer; Postoperative pain; Flurbiprofen axetil; Quality of life

Core Tip: This study demonstrates that electroacupuncture (EA) combined with flurbiprofen axetil (FA) provides superior analgesia and functional recovery after breast cancer surgery compared with FA alone. EA + FA reduced postoperative pain, shortened recovery time, and improved quality of life. These findings highlight EA as an effective non-pharmacological adjunct to enhance perioperative care and promote early rehabilitation in breast cancer patients.

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INTRODUCTION

Breast cancer has the highest worldwide incidence in women among all malignancies, with a rising incidence in recent years[1]. Surgical treatment is the most important part of the comprehensive treatment of breast cancer[2]. but postoperative pain is a common complication after breast cancer surgery[3]. Opioids and nonsteroidal drugs are commonly used in postoperative pain management[3], because of its side effects, it seriously affects the rapid recovery after surgery. Acupuncture has shown promise in the perioperative period, such as reducing postoperative pain[4]. Therefore, this study aims to explore the effects of electroacupuncture (EA) combined with flurbiprofen axetil (FA) on postoperative pain and early functional recovery of breast cancer patients, and provide reference for the promotion and application of acupuncture in perioperative medicine.

MATERIALS AND METHODS

Research participants

The research protocol was approved by the Ethics Committee of Guangdong Provincial Hospital of Chinese Medicine (Approval No. Ethics Committee of Guangdong Provincial Hospital of Chinese Medicine BF2018-006-01). Written, informed participant assent or certifier consent was obtained for all patients. This study selected patients between February 12, 2019 and January 10, 2020 in our hospital (The Second Affiliated Hospital of Guangzhou University of Chinese Medicine).

Inclusion criteria: (1) Undergoing elective breast cancer surgery; (2) Age 18-75 years; (3) Body mass index 18-30 kg/m^2; (4) American Society of Anesthesiologists class I-III; (5) Karnofsky Performance Status score ≥ 60; and (6) Voluntary signing of informed consent before participating in the study.

Exclusion criteria: (1) Contraindication for EA stimulation, including patients with local skin damage, infection, or electrophysiological devices implanted at the acupoints; (2) Communication disorders and inability to cooperate with researchers, such as those caused by language comprehension deficits and certain mental illnesses; (3) Unstable angina or myocardial infraction within 3 months and New York Heart Association functional class ≥ 3; (4) Blood pressure ≥ 180/100 mmHg during preoperative visits, in accordance with the World Health Organization-International Society of Hypertension guidelines; (5) History of peptic ulcer/bleeding/perforation; (6) Severe liver or kidney function abnormalities (liver dysfunction: More than twice the normal upper limit of any of alanine transaminase, aspartate transaminase, alkaline phosphatase, or total bilirubin; renal dysfunction: Creatinine clearance rate of < 30 mL/minute or blood creatinine > 177 μmol/L); (7) Known allergies to nonsteroidal drugs; and (8) Participation in other clinical trials within 3 months before selection for this study.

Criteria for dropout cases: (1) Serious adverse events (*e.g.*, second operation, death), complications, or certain physiological changes that rendered the participant unsuitable for continuing the tests; (2) Withdrawal from the study on their own, not due to curative-effect factors or at the request of a guardian; (3) Request to withdraw by the researchers; and (4) Being lost to follow-up.

Discontinuation criteria: (1) Not meeting either the inclusion or exclusion criteria; (2) Consumption of drugs other than the prescribed medication, meaning that the efficacy and safety of drugs could not be evaluated in the participant; and (3) Becoming unblinded during the study.

Randomization

Patients were randomly assigned (1:1) to EA group and control group. Randomization was performed by block. Blocks of random length are used through the computer. No stratification factors were used. Randomization was conducted by investigators who were not involved in anesthesia or research index recording. Acupuncturists, anesthesiologists, statisticians, data collectors, and record managers involved in this trial worked independently and performed only their designated tasks. Once the protocol was initiated, there was no private communication between any researchers.

Interventions

We preoperatively evaluated patients and assigned them their random numbers once we had selected them for the study. After patients were randomized, they received their corresponding interventions by grouping. The EA group was given electrical stimulation at the Zusanli and Sanyinjiao acupoints (Hanyi Acupuncture Needles, 0.18 mm × 20 mm; Changchun AIK Medical Devices Co. Ltd., Changchun, China). The skin at the selected acupoints was disinfected by alcohol wipe and then connected to an electrical stimulator (Huatuo Sdz-V Nerve and Muscle Stimulator; Suzhou Medical Appliance Factory, Suzhou, China) to provide 2/10 Hz sparse-dense wave stimulation. Stimulation intensity was based on the tolerance and degree of soreness of individual patients, but was adjusted within a standardized and predefined safe range. All procedures were delivered by trained acupuncturists following an identical protocol to enhance reproducibility. Patients in the control group received sham acupuncture; the acupuncture needles were connected to the electrical stimulator and used to create a shallow puncture 0.5 cm away from the Zusanli and Sanyinjiao acupoints, with no needle manipulation. Patients in both groups were given uniform general anesthesia 30 minutes afterward, induced with propofol (2 mg/kg) and sufentanil (0.5 μg/kg). We performed routine monitoring of all patients by electrocardiogram and for peripheral oxygen saturation, blood pressure, and bispectral index. Each patient underwent peripheral intravenous (intravenous injection) cannulation to infuse Ringer's solution at a rate of 8 mL/kg/hour. Vecuronium (0.1 mg/kg) was administered when muscle relaxants were needed. During surgery, we continued to deliver the abovementioned drugs to patients at the maintenance doses requested by the surgeons according to patients' vital signs and to maintain depth of anesthesia (bispectral index 50 ± 5). None of the patients received inhalational anesthesia. Electrocardiogram, heart rate, blood pressure, respiratory rate, peripheral oxygen saturation, end-tidal CO_2, and amount of intake and output were intraoperatively monitored for each patient. When suturing was complete, the anesthetic-infusion pump was stopped, and each patient received an intravenous injection of 4.48 mg tropisetron and 50 mg FA. If additional analgesia was required, rescue analgesics such as morphine (1-3 mg intravenous injection) or paracetamol (1000 mg intravenous injection) were administered based on pain severity [Visual Analog Scale (VAS) scores]. EA stimulation was simultaneously terminated. All patients were sent to the recovery room after surgery; once they met the standard for discharge from the recovery room, they were transferred to the wards. We assessed the postoperative quality of life of patients.

Modern evidence also supports the use of Zusanli and Sanyinjiao for postoperative analgesia. Clinical and neurophysiological studies have shown that stimulation of these acupoints modulates nociceptive signaling, enhances autonomic regulation, and promotes endogenous opioid release[5-9]. Randomized controlled trials further demonstrate that acupuncture at Zusanli and Sanyinjiao reduces postoperative pain and perioperative analgesic requirements[10,11]. Therefore, Zusanli and Sanyinjiao were selected based on both traditional theory and contemporary evidence.

To reduce performance and assessment bias, patients and postoperative outcome assessors were blinded to group allocation. Sham acupuncture was performed by inserting needles superficially near the true acupoints and connecting them to an inactive stimulator to mimic the appearance of EA. Assessors and data analysts were not involved in treatment delivery and remained blinded throughout the study. Acupuncturists could not be blinded due to the nature of the intervention but did not participate in postoperative assessments.

Outcomes

The primary outcome was the incidence of moderate-to-severe movement-evoked pain (VAS > 4) at 24 hours postoperatively, assessed by the intensity of pain during ipsilateral upper-arm movement or walking. Secondary outcomes included: (1) Incidence of moderate to severe pain during exercise 48 hours and 72 hours after surgery; (2) Incidence of moderate to severe pain at rest 24 hours, 48 hours and 72 hours after surgery; (3) VAS pain scores 24 hours, 48 hours and 72 hours after surgery; (4) 24 hours postoperative basic activities of daily living (BADL) score: Using the modified Barthel Index assessment form[12], It includes washing (5 points), getting dressed and undressed (10 points), moving the bed and chair (15 points), eating (10 points), going to the bathroom (10 points), going up and down stairs (10 points), walking on the ground (15 points), taking a shower (5 points), stool control (10 points), and urination control (10 points). The full score is 100. Total score ≤ 40, heavy dependence, all need others care; 41 to 60, moderate dependence, most need others care; 61-99, mild dependence, a few need others care; 100 points no dependence, no need for other people to care; and (5) Occurrence of postoperative complications.

Sample size calculations

In our pilot trial of 23 patients undergoing elective breast cancer surgery, we observed that 5 patients of 12 patients (41.6%) in the EA group reported exercise-induced moderate to severe pain 24 hours after surgery, compared to 7 of 11 patients (63.6%) in the control group, assuming α = 0.05 and β = 0.2. With a possible 20% dropout rate, the final sample size was calculated as 65 patients for each group.

Statistical analysis

The investigator reviewed and retrieved the case observation forms after the trial and sent them to a specially assigned person for data processing. The database was locked after verification and confirmation. Measurement data were presented as mean ± SD and median (interquartile range); count data were presented as n (%) of cases. We used SPSS version 17.0 (IBM Corp., Armonk, NY, United States) for statistical analysis in this study.

We first analyzed the baseline demographics and characteristics of both groups to examine how well the groups were balanced and their comparability. Therapeutic efficacies were then compared between the groups. We compared quantitative data, such as age, height, and weight, between the groups using an independent-sample t test. Between-group clinical efficacies were compared using t test and rank-sum test. All statistical tests were two-sided. P < 0.05 was considered to be statistically significant.

RESULTS

Patient demographics and characteristics

Patient baseline characteristics were important for comparisons between the two groups. Given that age, height, weight, comorbidities, Surgical procedure, anesthesia time, and intake and output were factors with potential effects on the quality of postoperative recovery of patients, we collected and recorded all possible demographic and characteristic data as a baseline to describe the homogeneity of both groups of breast cancer patients. 130 patients were included in the statistics. No significant differences in baseline characteristics were found between the two groups (P > 0.05; Table 1).

Primary outcome

EA reduced the incidence of moderate to severe pain during exercise 24 hours after surgery. Specifically, 32 of 65 (49.2%) in the EA group reported moderate to severe pain during exercise 24 hours after surgery (30 moderate and 2 severe), compared to 42 of 65 (64.6%) in the control group (41 moderate and 1 severe; P < 0.05; Table 2).

Secondary outcomes

Post-operation pain: Compared with the control group, EA did not reduce the incidence of moderate to severe pain during exercise at 48 hours and 72 hours after surgery (P > 0.05). There was no significant difference in the incidence of moderate to severe pain during rest at 24 hours, 48 hours, and 72 hours after surgery (P > 0.05). In addition, the VAS score of pain both exercise and rest at 24 hours after surgery was decreased by EA (P < 0.01). However, there was no significant difference in VAS scores of exercises and rest pain between the two groups at 48 hours and 72 hours after surgery (P > 0.05; Table 2).

BADL scores 24 hours post-surgery: BADL scores and degree for bathing, grooming, dressing, toileting, and transferring (moving in and out of a bed or chair) in the EA group were significantly better than in the control group at 24 hours post-surgery (P < 0.05). The total BADL score of the EA group at 24 hours post-surgery was significantly higher than that of the control group (P < 0.01). Data are shown in Table 3.

Postoperative complications: Compared with the control group, EA can significantly reduce the incidence of postoperative nausea and vomiting (PONV) and improve postoperative appetite (P < 0.05; Table 4).

DISCUSSION

In our study, FA was used as basic postoperative analgesic therapy in both groups. Our results showed that EA combined with FA significantly reduced the incidence of moderate-to-severe pain during exercise and VAS scores at 24 hours postoperative.

From a traditional Chinese medicine perspective, postoperative pain is primarily caused by Qi stagnation and blood deficiency. Zusanli, a He-sea point of the stomach meridian, invigorates the spleen and stomach, while Sanyinjiao, located at the intersection of the three Yin channels, nourishes the liver, spleen, and kidney Qi. Their combined use enhances Qi and blood circulation, addressing the root causes of pain and promoting early recovery, which aligns with our observed improvements in movement-induced pain and functional recovery. EA therapy combines traditional acupuncture with electrical stimulation, providing continuous stimulation to specific acupoints. This approach induces morphological and functional changes in target organs and tissues, contributing to the analgesic effects[8]. In addition, low-frequency and high-frequency electrical stimulation have distinct effects on specific brain regions involved in pain modulation. Low-frequency stimulation has been shown to activate the hypothalamic arcuate nucleus, while high-frequency stimulation primarily targets the pontine parabrachial nucleus, both of which play key roles in pain perception and autonomic regulation. The alternating stimulation of these regions promotes the simultaneous release of enkephalins and dynorphins, endogenous opioid peptides that bind to opioid receptors in the central nervous system, effectively lowering the pain threshold and enhancing analgesic effects[6]. This combined mechanism of neural modulation is thought to

Table 1 Patients demographic characteristics in two groups, median (interquartile range)/*n* (%)

Demographic characteristics	Control group (*n* = 65)	EA group (*n* = 65)
Year	52 (43.5-61)	52 (43.5-61.5)
Height (cm)	156 (151.5-160)	155 (152-159)
Weight (kg)	55 (50-60.25)	57 (51.25-67.75)
Combined disease	29 (44.60)	36 (55.40)
Breast cancer location		
Right	29 (44.60)	33 (50.80)
Left	36 (55.40)	32 (49.20)
Basic information of operation		
Incoming (mL)	1000 (1000-1000)	1000 (1000-1000)
Output (mL)	120 (30-230)	110 (20-250)
Anesthesia time (minutes)	240 (190-285)	240 (189.5-277.5)
Operation procedure		
Partial resection + sentinel lymph node biopsy	7 (5.38)	8 (6.16)
Partial resection + axillary dissection	9 (6.92)	8 (6.16)
Total mastectomy + sentinel lymph node biopsy	36 (27.69)	30 (23.08)
Total mastectomy + axillary dissection	18 (13.85)	14 (10.76)

EA: Electroacupuncture.

Table 2 Occurrence of postoperative pain in two groups

Time after surgery	*n* (moderate/severe pain) (%)		VAS score (mean ± SD)	
	Control group	EA group	Control group	EA group
24 hours after surgery				
Pain during exercise	41/1 (64.6)	30/2 (49.2)[a]	3.954 ± 0.233	2.862 ± 0.233[b]
Pain at rest	11/1 (18.5)	7/1 (12.3)	2.692 ± 0.174	1.789 ± 0.174[b]
48 hours after surgery				
Pain during exercise	29/0 (44.6)	24/1 (38.5)	2.108 ± 0.18	1.785 ± 0.18
Pain at rest	6/0 (9.2)	5/0 (7.7)	1.169 ± 0.119	0.631 ± 0.119
72 hours after surgery				
Pain during exercise	19/0 (29.2)	16/0 (24.6)	1.215 ± 0.12	1.115 ± 0.12
Pain at rest	2/0 (3.1)	3/0 (4.6)	0.631 ± 0.08	0.415 ± 0.086

[a]P < 0.05 *vs* control group.
[b]P < 0.01 *vs* control group.
EA: Electroacupuncture.

contribute to the observed improvements in postoperative pain relief and functional recovery. Moreover, the activation of these brain regions can also influence autonomic functions, including heart rate and blood pressure regulation, which further supports the role of EA in promoting recovery beyond pain management.

Functional magnetic resonance imaging has also been used to assess the effects of acupuncture on brain connectivity networks, with task-positive network activation and task-invisible network inactivation observed during acupuncture, thus modifying pain[7,9]. Moreover, according to the specificity of acupuncture, the brain has different regulation of true and false acupoint response[5]. Therefore, in this study, 2/100 Hz EA stimulation was selected, sham acupuncture (false acupoint) was used as a control, and combined with FA was used to play a better analgesic effect. At present, relevant clinical studies have confirmed that the combination of both can play a synergistic role in analgesia[10,11], the results are similar with our study. But our study also found that EA combined with FA was more effective at relieving exercise-

Table 3 Basic living ability scores 24 hours post-surgery in two groups, median (interquartile range)

BADL item	Control group (*n* = 65)	EA group (*n* = 65)
Eating	10 (10-10)	10 (10-10)
Bathing	0 (0-2.5)	0 (0-5)
Grooming	0 (0-5)	5 (5-5)[b]
Dressing	5 (5-10)	10 (5-10)[b]
Controlling defecation	10 (5-10)	10 (10-10)
Controlling urination	10 (5-10)	10 (10-10)
Toileting	10 (5-10)	10 (10-10)[a]
Transferring	15 (10-15)	15 (10-15)[a]
Walking	15 (15-15)	15 (15-15)
Climbing up and down the stairs	5 (0-5)	5 (0-10)
Total score	80 (65-87.5)	85 (75-90)[b]

[a]$P < 0.05$ *vs* control group.
[b]$P < 0.01$ *vs* control group.
BADL: Basic activities of daily living; EA: Electroacupuncture.

Table 4 Postoperative complications in two groups, *n* (%)

Complication	Control group	EA group
Dizziness	5 (7.8)	4 (6.2)
Appetite loss	36 (55.4)	27 (41.5)[a]
Postoperative nausea and vomiting	22 (33.8)	13 (20.0)[a]
Diarrhoea	4 (6.2)	4 (6.2)
Constipation	4 (6.2)	4 (6.2)

[a]$P < 0.05$ *vs* control group.
EA: Electroacupuncture.

induced pain, especially if the pain is moderate. This was rarely reported in previous studies.

The results of our study also showed no difference in the incidence of moderate to severe pain (including resting pain and exercise pain) and VAS scores between the two groups at 48 hours and 72 hours after surgery. This may be related to the timeliness of acupuncture. After EA stimulation, acupuncture effect does not exist persistently, that is time-efficient [13]. There is a brief description of the frequency of acupuncture in “one day”, “two days”, and “Long sick, evil into the deep...and from time to time, they stab again” in *Linshu - Zhongsh*. It is suggested that the frequency of acupuncture treatment should be selected according to the kinds of diseases.

Because exercise-induced pain interferes normal daily activities, such as dressing, grooming, toileting and walking, the quality of life after operation is affected. Moderate to severe pain induced by daily activities after breast cancer surgery often reduces the pain by restricting the movement of the affected upper arm, which results in impaired functional recovery after surgery. Therefore, the improvement in BADL scores at 24 hours after surgery, including grooming, dressing, toilet use, and bed chair movement, may be due to the relief of exercise-induced pain, reflecting enhanced early functional recovery. At the same time, EA can reduce the occurrence of postoperative complications (nausea, vomiting and loss of appetite). As we all know, acupuncture is a safe, non-drug measure with no adverse reactions. Fourth Consensus Guidelines for the Management of PONV is clearly suggested that acupuncture has the same effect as any of the 6 antinausea drugs commonly used in clinical prevention and treatment of PONV, and the effect is better than false acupuncture[14]. Our study is also same. Given its low cost, minimal risk, and non-pharmacological characteristics, EA is a potentially cost-effective adjunct to multimodal analgesia, although formal economic evaluation was beyond the scope of this study. Incorporating EA may help reduce functional impairment and improve short-term postoperative recovery in a resource-efficient manner.

EA has good efficacy and safety in this study, but there are still some limitations: (1) Although a standardized sham procedure was applied, complete blinding in acupuncture trials is inherently challenging. Acupuncturists could not be blinded, and subtle differences in sensation might have been detected by some patients, potentially introducing expectation bias. However, the use of sham stimulation and blinded outcome assessors minimized the risk of such bias;

(2) In our study, the time window of EA treatment was only limited to 30 minutes before surgery to the end of surgery, which may be the reason why exercise-induced pain could only be relieved within 24 hours, while there was no significant difference in pain at 48 hours and 72 hours after surgery; (3) Postoperative acupuncture is not continued, which may reduce the analgesic effect. However, it provides a novel postoperative analgesia strategy that incorporates EA into postoperative multimodal analgesia, including postoperative acute and chronic pain. Therefore, it is necessary to conduct a comprehensive study on this issue in the future; (4) This study assessed only short-term outcomes and lacked long-term follow-up; therefore, the potential effects of perioperative acupuncture on chronic postoperative pain and sustained functional recovery remain unknown. In addition, rescue opioid consumption was not recorded or compared between groups, limiting the ability to quantify the incremental analgesic effect of EA; and (5) Patient-reported satisfaction is an important indicator of the acceptability of traditional medicine interventions, but was not evaluated in this study. Moreover, the BADL assessment at 24 hours may not fully reflect functional mobility in the acute inpatient setting, and future studies should consider using scales validated for early postoperative recovery.

CONCLUSION

In summary, EA combined with FA can significantly reduce the incidence of moderate to severe pain during exercise 24 hours after surgery, reduce pain VAS score 24 hours after surgery, reduce postoperative complications, and improve early functional recovery 24 hours after surgery.

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FOOTNOTES

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Author contributions: He BM did the statistical analyses; He BM and Hong QX initiated the project, generated research funds and ideas, led and coordinated the project, interpreted data, and wrote the paper; Sun XL provided intellectual input; Sun XL and Huang XY initiated and performed the experiments; Huang XY and Zhang SS were responsible for patients' inclusion, follow-up, and adjudication of outcomes. All authors have read and agreed to the published version of the manuscript.

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